Appendiceal Neoplasms

Classification of Appendiceal Neoplasms

Serrated Lesions and Polyps

Mucinous Neoplasms

- Low-grade Appendiceal Mucinous Neoplasms (LAMN)
- High-grade Appendiceal Mucinous Neoplasms (HAMN)

Adenocarcinoma

- Mucinous (>50%)
- Non-mucinous (<50%)

Goblet Cell Adenocarcinoma

Neuroendocrine Neoplasms

- well-differentiated NETs
- poorly differentiated NECs
- MiNENs

A. Serrated Lesions and Polyps Analogous to colorectal serrated lesions; confined to the mucosa.

Cured by complete excision (appendectomy).

B. Mucinous Neoplasms Low-Grade Appendiceal Mucinous Neoplasm (LAMN):

The most common appendiceal epithelial neoplasm (~0.3% of appendix specimens). [12]

Characterized by tall mucinous epithelium with a pushing border (not infiltrative glands with desmoplasia). [10]

Associated with atrophy of the appendiceal wall, loss of lymphoid tissue, and effacement of the muscularis mucosa.

Mucin pools can extend into the serosa and beyond.

Molecular profile: almost always MSS, BRAF V600E negative, frequently harbor KRAS and GNAS mutations. [10][13]

The most common cause of pseudomyxoma peritonei (PMP), accounting for up to 94% of cases. [12]

HIPC

High-Grade Appendiceal Mucinous Neoplasm (HAMN):

Retains features of LAMN but with high-grade cytologic features and complex architecture. [10]

Staged as mucinous adenocarcinoma; data are limited.

May additionally harbor TP53 and ATM mutations not seen in LAMN.

C. Adenocarcinoma (AA) Rare outside of LAMN/PMP; can arise from mucinous neoplasms, intestinal-type adenomas, or GCAs. [10]

Distinguished from mucinous neoplasms by: high cellularity, high-grade nuclear features, infiltrative growth pattern with desmoplastic response. [10]

Classified as mucinous (>50% mucin) or non-mucinous (50% mucin). [10]

Signet ring cell carcinoma (>50% signet ring cells) carries the worst prognosis. [10]

Non-mucinous subtypes may be MSI-H; mucinous subtypes are rarely MSI-H. [10]

D. Goblet Cell Adenocarcinoma (GCA) Previously classified as a NET ("goblet cell carcinoid") but now recognized as a distinct entity with biologic behavior more similar to adenocarcinoma. [10]

Characterized by deep mucosal/submucosal circumferential infiltration without surface epithelial dysplasia. [10]

Grading is based on percentage of low-grade (tubular/clustered) component:

Grade 1: >75% low-grade pattern

Grade 2: 50–75% low-grade pattern

Grade 3: <50% low-grade pattern [10]

Molecular profile: lower rates of KRAS and APC mutations; MSI and TMB-H are rare. [10]

Even low-grade GCA can present with metastatic disease due to deeply invasive nature. [10]

E. Neuroendocrine Neoplasms (NENs) The most common appendiceal neoplasm overall (~52% of incidental tumors). [1][15]

Includes well-differentiated NETs, poorly differentiated NECs, and MiNENs.

~80% are <1 cm at diagnosis. [16]

Managed per NCCN Neuroendocrine and Adrenal Tumor guidelines. [10]